

Cohort Exercise - Use Case

Learning Outcomes

By the end of this exercise, you should be able to: 1. Analyse system requirements using use cases and use case diagrams.

Total 10 Marks

Note: The total score calculation is represented as $\sum_{i=1}^3 Task_i = 10$ marks.

Task 1 (3 Marks)

Consider the following user requirements of a campus food ordering app.

A student signs up for an account in the campus food ordering system. During the sign-up, the student needs to supply the following particulars: 1. the student's email, preferred password, and mobile number. 2. the student's campus delivery details, such as hostel room number or classroom location. 3. the preferred dietary requirements (e.g., Halal, Vegetarian).

Upon successful sign-up, the system sends a confirmation email to the student.

After login, the student browses through the list of food courts inside the campus. The student selects a preferred food court stall to view its available menu items. Upon selecting an item and specifying the delivery location (hostel or classroom), the student needs to pay for the meal that she orders using a credit card. The system verifies the credit card payment.

Upon successful payment, the system sends an email confirmation to the student for her order. The system sends an email and SMS notification to the food court stall regarding the new order. When the payment is unsuccessful, the system prompts the student to use a different payment method. The student can log out from the system anytime she wants.

Analyse and document the above system requirement in at least **two** use case descriptions.

Task 2 (4 Marks)

Continue from the system requirements from Task 1.

Ten minutes before the estimated delivery time, the system sends an SMS reminder to the student and the delivery rider about the upcoming delivery to the specified classroom or hostel. The food court stall updates the order status to "Preparing", and the rider picks up the order. The student tracks the meal delivery using the student's account.

When the delivery ends, the app prompts the student to leave a food review. The student writes a review and submits the rating to the food court stall through

the system. The stall owner acknowledges the student's review and gives a reply and a comment.

Analyse and document the above system requirement in use case descriptions. (You decide how many are there.)

Task 3 (3 Marks)

Summarize the use cases from Task 1 and Task 2 in a use case diagram. You are encouraged to use coding languages such as plantuml, mermaid.

Assignment AI Usage Policy

This assignment evaluates both your conceptual systems design skills and your ability to generate technical diagrams. To support your learning, the use of Generative AI (e.g., ChatGPT, Gemini, Claude) is governed by the following strict rules:

- **Task 1 & Task 2 (Use Case Descriptions): NO AI ALLOWED.** The core logic of your system must be 100% your own work.
- **Task 3 (Use Case Diagram): AI ASSISTANCE PERMITTED.** You are allowed to use AI tools to translate your manually written use cases from Tasks 1 and 2 into **PlantUML** or **Mermaid** code. The AI should act as your syntax generator, not your system designer.
- **Prompt Transparency Requirement.** If you use AI to help generate your **Task 3** code, you **must** include a short section at the end of your document containing the exact prompts you used to generate and refine the diagram. Work submitted with AI-generated code but missing the prompt history may face academic integrity review.
- **No Penalty for Manual Coding.** Using AI is entirely optional. There is absolutely no penalty (0 marks deducted) if you prefer to write your **PlantUML** or **Mermaid** code entirely by hand without utilizing AI.
